DEEPIKA PIRIYA

AI Engineer | Generative AI | Agentic AI | Machine Learning | RAG
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PROFESSIONAL SUMMARY

AI and analytics professional with hands-on project experience in Generative AI, LLMs, RAG, multi-agent workflows, NLP, machine learning, and recommendation systems. Builds Python-based AI applications using OpenAI API, LangChain, ChromaDB, FastAPI, and Docker, alongside professional experience in SQL analytics, Power BI, ETL, data quality, KPI reporting, and workflow automation.

TECHNICAL SKILLS

Programming & Data Science: Python, SQL, Pandas, NumPy, EDA, Statistical Analysis, Data Cleaning, Data Validation
AI & Generative AI: Agentic AI, Multi-Agent Systems, LLMs, Generative AI, RAG, Vector Databases, Embeddings, Semantic Search, Retrieval Optimization, Prompt Engineering, Prompt Templates, AI Orchestration, AI Application Development, NLP
LLM & Machine Learning Engineering: LangChain, Hugging Face, Transformers, Scikit-learn, Feature Engineering, Model Evaluation, LLM Evaluation, Machine Learning Pipelines, Recommendation Systems, SVD, Cosine Similarity, Model Deployment
Development & MLOps: FastAPI, Flask, REST APIs, Docker, Streamlit, Gradio, Git, CI/CD, MLOps, Experiment Tracking
Cloud & Data Platforms: AWS, Azure OpenAI, Azure AI Search, GCP, Azure Data Factory, ChromaDB
Analytics & Data Engineering: Power BI, Power Query, DAX, Business Intelligence, KPI Development, ETL, Data Warehousing, Data Governance, Root Cause Analysis, Stakeholder Management, Data Modeling, SQL CTEs, Window Functions
Additional ML Concepts: Fine-Tuning, Feature Stores

PROFESSIONAL EXPERIENCE

JMBAXI Visakhapatnam
Data Analytics Engineer - Trainee BD Jan 2025 – Feb 2026

- Developed 6+ Power BI dashboards using star-schema data models to track business KPIs, reducing report turnaround time by 35%.
- Optimized SQL queries using CTEs, window functions, and joins across 500K+ records to improve data retrieval and analytical reporting.
- Performed data cleaning, transformation, and validation using Power Query, maintaining 99%+ data accuracy for reliable analysis.
- Automated 8+ recurring reports, saving approximately 6 hours per week and improving reporting efficiency.
- Applied data analysis and root cause analysis techniques to identify data inconsistencies, uncover trends, and support data-driven decision-making.
- Applied Python, SQL, and data preprocessing concepts relevant to machine learning workflows and AI-ready data preparation.

AI AND ML PROJECTS

Multi-Agent Customer Support and Knowledge Agent
Python, LangChain, RAG, OpenAI API, ChromaDB, FastAPI, Docker

- Built a multi-agent system for query classification, knowledge retrieval, response generation, and escalation.
- Deployed via FastAPI and containerized with Docker for production-ready access.

Conversational RAG Q&A Agent
Python, OpenAI API, LangChain, ChromaDB, FastAPI

- Developed a RAG-based QA system over 500+ unstructured PDFs using embeddings and vector search.
- Improved retrieval accuracy through optimized text chunking and prompt engineering.

Machine Learning Prediction System
Python, Scikit-learn, Machine Learning, Model Evaluation

- Built and evaluated supervised ML models with feature engineering and performance tracking.

EDUCATION

MBA, Business Analytics, Andhra University 2023 – 2025
BCA, Adikavi Nannaya University 2020 – 2023

CERTIFICATIONS

- Data Science with GenAI – FLM
- Machine Learning with Python – IBM Cognitive Class
- Power BI and AI – Coval University